Elijah Verhoff

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Education

University of Maine – Bachelor of Science in Mechanical Engineering, Robotics Focus — May 2026

Experience

Project Engineer (Contract), The ONIX Corporation – Remote — Jan 2026 – Jun 2026

- Prevented over $200,000 in incorrect purchases by validating vendor quotations against 2D CAD, 3D assembly models, and P&ID documentation to catch specification mismatches and over-ordering
- Cut per-unit cost of a new drag-chain conveyor by ~$5,000 through design-for-manufacturability changes, replacing welded angle-iron rails with bolted joints and minimizing bent sections
- Ensured motors, actuators, sensors, pneumatics, and drives were appropriately NFPA and NEMA rated for rotary airlocks handling explosive fine particulate in hazardous-dust environments
- Reviewed control schematics, electrical drawings, and PLC/HMI logic and revised 3D CAD models and BOMs to keep mechanical and controls scope on specification

Engineering Specialist, Advanced Structures and Composites Center – Orono, ME — Dec 2024 – Dec 2025

- Led development of an in-situ winding mechanism for a continuous forming machine, sizing the rotating assembly and frame from Mathcad analysis of roller displacement, critical velocity, and structural reactions
- Designed a custom slip-ring and winder assembly in SolidWorks that delivered embedded power and tension-transducer feedback for closed-loop web-tension control
- Reduced material use 22% on large-area additive manufacturing by tuning overhang lengths and authoring ASTM-informed work instructions for thermoplastic strengthening tests

Student Research Assistant, Advanced Manufacturing Center – Orono, ME — June 2018 – Sept 2018

- Produced fully dimensioned SolidWorks drawing packages with GD&T to support manufacturing, inspection, and reverse engineering of mechanical components
- Generated CAM programs and operated 5-axis CNC mills, lathes, and laser cutters to fabricate precision parts

Projects

Rapidly Deployable Inflatable Hallway Barrier (Capstone) — elijahverhoff.github.io/projects/capstone

- Led controls and systems integration for a Sandia National Laboratories sponsored barrier that seals indoor choke points on demand
- Achieved full deployment in 15 seconds at a 50 psi supply through Arduino-driven pneumatic automation with pressure-feedback sensing and diagnostic logic

Stepper Motor Choir — elijahverhoff.github.io/projects/stepperchoir

- Built an electromechanical instrument that performs four-part harmony on synchronized stepper motors, with a Python pipeline parsing MusicXML into motor-motion commands
- Implemented Arduino firmware for coordinated multi-axis motor control and real-time playback of polyphonic scores

Skills

Mechanical Design: SolidWorks, Rhinoceros 3D, Fusion 360, GD&T, design for manufacturing
Engineering Analysis: Abaqus, MATLAB, Python
Robotics and Controls: ROS2, robot kinematics, motion planning, vision systems, PLC/HMI logic, closed-loop control
Manufacturing: CNC machining, CAM programming, prototyping, additive manufacturing